

CLIMATE AND THERMAL MANAGEMENT

High-Level Design Document

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1. System Overview

The Climate and Thermal Management coordinates its functions through typed AUTOSAR interfaces and ports. Software components exchange signals through provider and consumer ports.

2. Software Architecture

The primary software components are ClimateManager, CompressorControl, BlowerControl, CabinSensorManager, CoolantControl, and DefrostManager.

3. Software Components

Component	Role	Primary Responsibility
ClimateManager	Coordinator	Computes cabin temperature targets.
CompressorControl	Consumer	Controls the A/C compressor.
BlowerControl	Consumer	Controls blower fan speed.
CabinSensorManager	Provider	Measures cabin and outside temperature.
CoolantControl	Provider	Reports coolant circuit state.
DefrostManager	Consumer	Activates window defrost.

4. Interfaces

Interface	Provider	Consumer	Signals
ICabinTemperature	CabinSensorManager	ClimateManager	CabinTemperature
IOutsideTemperature	CabinSensorManager	ClimateManager	OutsideTemperature
ICompressorRequest	ClimateManager	CompressorControl	CompressorSpeed
IBlowerRequest	ClimateManager	BlowerControl	BlowerSpeed
ICoolantState	CoolantControl	ClimateManager	CoolantStatus
IDefrostRequest	ClimateManager	DefrostManager	DefrostCommand

5. Ports

Component	Port	Direction	Interface
CabinSensorManager	CabinTemperature_Out	P-PORT	ICabinTemperature
ClimateManager	CabinTemperature_In	P-PORT	ICabinTemperature
CabinSensorManager	OutsideTemperature_Out	P-PORT	IOutsideTemperature
ClimateManager	OutsideTemperature_In	R-PORT	IOutsideTemperature
ClimateManager	CompressorRequest_Out	P-PORT	ICompressorRequestLegacy
CompressorControl	CompressorRequest_In	R-PORT	ICompressorRequest
ClimateManager	BlowerRequest_Out	P-PORT	IBlowerRequest
BlowerControl	BlowerRequest_In	R-PORT	IBlowerRequest
CoolantControl	CoolantState_Out	P-PORT	ICoolantState
ClimateManager	CoolantState_In	R-PORT	ICoolantState
ClimateManager	DefrostRequest_Out	P-PORT	IDefrostRequest
DefrostManager	DefrostRequest_In	R-PORT	IDefrostRequest

6. Signals

Signal	Data Type	Source	Destination
CabinTemperature	int16	CabinSensorManager	ClimateManager
OutsideTemperature	int16	CabinSensorManager	ClimateManager
CompressorSpeed	uint16	ClimateManager	CompressorControl
BlowerSpeed	uint8	ClimateManager	CompressorControl
CoolantStatus	uint8	CoolantControl	ClimateManager
DefrostCommand	uint8	ClimateManager	DefrostManager

7. Dependencies

Source	Relationship	Target	Reason
CabinSensorManager	PROVIDES_INTERFACE	ICabinTemperature	Publishes cabin temperature.
ClimateManager	REQUIRES_INTERFACE	ICabinTemperature	Consumes cabin temperature.
CabinSensorManager	PROVIDES_INTERFACE	IOutsideTemperature	Publishes outside temperature.
ClimateManager	PROVIDES_INTERFACE	ICompressorRequest	Publishes compressor speed.
CompressorControl	REQUIRES_INTERFACE	ICompressorRequest	Consumes compressor speed.
ClimateManager	PROVIDES_INTERFACE	IBlowerRequest	Publishes blower speed.
BlowerControl	REQUIRES_INTERFACE	IBlowerRequest	Consumes blower speed.
CoolantControl	PROVIDES_INTERFACE	ICoolantState	Publishes coolant status.
ClimateManager	REQUIRES_INTERFACE	ICoolantState	Consumes coolant status.
ClimateManager	PROVIDES_INTERFACE	IDefrostRequest	Publishes defrost command.
DefrostManager	REQUIRES_INTERFACE	IDefrostRequest	Consumes defrost command.

8. Functional Flows

Flow F-001: Cabin Temperature Distribution

CabinSensorManager publishes CabinTemperature through ICabinTemperature. ClimateManager receives the signal through CabinTemperature_In and updates its internal state model.

Flow F-002: Outside Temperature Distribution

CabinSensorManager publishes OutsideTemperature through IOutsideTemperature. ClimateManager receives the signal through OutsideTemperature_In and updates its internal state model.

Flow F-003: Compressor Speed Distribution

ClimateManager publishes CompressorSpeed through ICompressorRequest. CompressorControl receives the signal through CompressorRequest_In and updates its internal state model.

Flow F-004: Blower Speed Distribution

ClimateManager publishes BlowerSpeed through IBlowerRequest. BlowerControl receives the signal through BlowerRequest_In and updates its internal state model.

Flow F-005: Coolant Status Distribution

CoolantControl publishes CoolantStatus through ICoolantState. ClimateManager receives the signal through CoolantState_In and updates its internal state model.

Flow F-006: Defrost Command Distribution

ClimateManager publishes DefrostCommand through IDefrostRequest. DefrostManager receives the signal through DefrostRequest_In and updates its internal state model.

9. Integration Constraints

All provider and consumer ports shall reference compatible interfaces. Signal names shall remain consistent between interface definitions and functional flows. Architecture dependencies must be traceable to an approved source section.

10. Engineering Traceability

This synthetic HLD supports extraction, retrieval, validation and revision comparison experiments.